

Revenue Leakage Investigation

Project Report

FreshMart Retail Pvt. Ltd. (fictional)

Project question

Are inventory stockouts and poor product availability contributing to missed sales, and where should FreshMart focus its attention?

This project uses one retail inventory dataset and follows the problem from data quality checks through Python analysis, SQL analysis, Power BI reporting, findings and practical recommendations.

The report is written as a record of the analysis work completed for the portfolio project. It focuses on what was actually done and what the data can reasonably support.

1. Executive Summary

The project investigates a simple business problem: products cannot be sold when they are not available. The analysis therefore looks at stockouts, demand gaps, inventory utilization, overstock and sales performance to understand where availability problems are occurring.

The dataset contains 73,100 daily records across 5 stores, 20 Product IDs, 5 categories and 4 regions. After data-quality checks and feature engineering, the final analysis dataset still contains 73,100 records but has 24 columns.

The analysis found 2,585 records marked as stockouts. Stockouts are spread across all categories and regions, while some stores show higher stockout counts than others. S004 has the highest store-level stockout count at 547.

Toys has the highest category-level demand gap at 74,913.95 units. The Power BI analysis also estimates an overall revenue opportunity of about 26.88M currency units. This is an estimate based on the project calculations, not a guaranteed amount of recoverable revenue.

Main takeaway

The problem does not appear to be limited to one category. The stronger operational signal is the combination of widespread stockouts and demand gaps, with S004, S005 and S002 standing out at store level.

2. Business Problem and Objective

FreshMart wants to understand whether inventory availability is affecting sales. A product can have customer demand but still record fewer sales when the available stock is not enough. That creates a possible missed-sales opportunity.

The project was built around five practical questions:

- Where are stockouts happening most often?
- Which stores and categories are most affected?
- Where is forecast demand higher than units sold?
- Which areas show stronger revenue opportunity?
- What actions could help reduce missed sales?

The goal was not to prove a causal relationship from the dataset alone. The goal was to find useful patterns, identify priority areas and turn those patterns into practical business actions.

3. Dataset Overview

Item	Value
Original records	73,100
Original columns	15
Date coverage	731 dates; 2022-01-01 to 2024-01-01
Stores	5
Product IDs	20
Categories	5
Regions	4
Final analysis columns	24

The source dataset is a retail inventory dataset with daily information about stores, products, sales, inventory, demand forecasts, pricing and related factors. The dataset source and license have been verified.
Dataset source: <https://www.kaggle.com/datasets/anirudhchauhan/retail-store-inventory-forecasting-dataset>
License: CC0: Public Domain (CC0 1.0 Universal)

Currency note
The source dataset does not disclose a currency unit. Monetary values in the final project should therefore be presented as currency units, without assuming INR, USD or another specific currency.

4. Data Validation and Cleaning

Before building calculations or dashboards, the raw data was checked for common quality problems. The checks covered missing values, duplicate records, date values, store and product identifiers, categories, regions and numeric fields.

Check	Result / action
Rows and columns	73,100 rows and 15 columns
Missing values	No general missing-value issue found in the raw dataset
Complete duplicates	No complete duplicate rows found
Date + Store + Product duplicates	No duplicate combinations found
Date range	2022-01-01 to 2024-01-01
Negative demand forecasts	673 records found and changed to 0
Discount	Values fall between 0 and 20
Holiday/Promotion	Stored as 0/1 values

One notable issue was 673 negative Demand Forecast values. These were changed to 0 before further analysis. The decision was documented rather than leaving negative forecast values in the business calculations.

The final feature-engineered file also contains 673 blank values in Fulfillment Rate (%). This is recorded as a characteristic of the final file and should not be silently treated as a business result.

5. Feature Engineering

The original fields were enough for basic exploration, but the business question needed a few calculated measures. Feature engineering was used to create those fields before the SQL and Power BI stages.

New field	How it is used
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New field	How it is used
Final price	Price after applying the discount
Revenue	Units Sold × Final price
Remaining inventory	Inventory Level – Units Sold
Inventory utilization	Measures how much available inventory was used
Demand gap	Demand Forecast – Units Sold
Fulfillment rate	Compares units sold with forecast demand, with zero-forecast handling
Stockout	Flags records where forecast demand is greater than available inventory
Overstock	Identifies records meeting the project's overstock condition
Overstock flag	Stores the overstock result as a usable category/flag

Important interpretation

Demand gap is a calculated measure, not a direct count of lost sales. A positive gap means forecast demand is higher than units sold. The project uses stockout records alongside this gap when discussing possible missed sales.

6. Exploratory Data Analysis

Python EDA was used to understand the shape of the data before moving into the dashboard. The analysis covered revenue, units sold, inventory, price, discount, demand forecast, demand gap, fulfillment, overstock and stockout patterns.

Area	Observation
Revenue	Total recorded revenue was about 494.97M currency units
Units sold	Average units sold per record was about 136.46
Inventory	Average inventory level was about 274.47 units
Demand forecast	Average forecast was about 141.53 units
Stockouts	2,585 records, about 3.54% of all records
Overstock	59,037 records, about 80.76% of all records
Demand gap	48,783 positive, 157 zero and 24,160 negative records
Price vs revenue	Positive correlation of about 0.469
Discount vs units sold	Very weak correlation, about 0.003
Inventory vs units sold	Positive correlation, about 0.590

The EDA also showed that revenue is fairly close across categories. Furniture is the highest-revenue category at about 100.23M currency units, while Electronics is lowest at about 97.39M.

A useful finding was the positive relationship between inventory level and units sold. This does not prove that more inventory causes more sales, but it supports looking at availability when investigating sales performance.

The EDA also showed that stockouts are a minority of records rather than the majority. This is important because the project is investigating the impact of affected records, not claiming that every record is a stockout.

7. SQL Analysis

SQL was used to move from general exploration to business-focused questions. The analysis was organized around revenue, stockouts, demand gaps, inventory utilization, overstock, rankings and stockout impact.

SQL analysis	Key result
Total revenue	494,971,391.76 currency units
Highest-revenue region	East — 124,792,224.91
Highest-revenue store	S005 — 100,503,643.04
Highest-stockout region	West — 673
Highest-stockout store	S004 — 547
Highest-stockout category	Groceries — 535
Highest-demand-gap region	South — 94,132.02
Highest-demand-gap store	S005 — 75,416.22
Highest-demand-gap category	Toys — 74,913.95

The SQL work also included store- and product-level comparisons against average performance, ranking queries using RANK(), fulfillment-rate comparisons and analysis of revenue and demand gaps specifically during stockout records.

The final SQL file was refined so that comparisons are made at the same level as the question. For example, store revenue is first aggregated to store level before comparing stores with the average store revenue, and product demand gaps are aggregated to product level before the comparison.

SQL result worth highlighting

S004 has the highest stockout count at 547 and also the highest demand gap during stockout records at 7,225.39. This is a useful pattern for operational investigation, but it should not be described as proof that stockouts caused the demand gap.

8. Power BI Dashboard

Power BI was used to turn the analysis into an interactive report. The dashboard has three pages, each answering a different business question.

Page	Purpose	Main content
Page 1 — Executive Overview	What is happening?	Revenue, units sold, demand gap, stockouts, revenue opportunity, revenue trend and category/store views
Page 2 — Inventory Performance Analysis	Where is the problem?	Stockouts, demand gap, overstock, lost units due to stockouts, category and product-level views
Page 3 — Insights & Recommendations	What should FreshMart do next?	Key findings, recommendations, expected impact and executive conclusion

The dashboard includes filters for date, region, category and store. The revenue trend also supports different time views so the same visual can be used to look at the data at different levels.

The dashboard was reviewed for consistency and usability. The revenue trend was corrected to display months in chronological order, and a duplicated demand-gap visual on Page 2 was replaced with an overstock view.

9. Key Findings

9.1 Stockouts are spread across the business

Groceries had the highest stockout count at 535, followed by Furniture at 524, Clothing at 510, and Toys and Electronics at 508 each. The small difference between categories suggests that the problem is not isolated to one category.

9.2 Some stores deserve earlier attention

S004 had the highest stockout count at 547, followed by S005 at 527 and S002 at 523. These three stores are useful starting points for a closer operational review.

9.3 Demand gaps are also broad

Toys had the highest category demand gap at 74,913.95 units. Furniture and Clothing were close behind. At store level, S005 had the highest total demand gap at 75,416.22 units.

9.4 The revenue opportunity is larger than one category

The Power BI analysis estimates about 26.88M currency units of overall revenue opportunity. The strongest category opportunities are spread across Toys, Clothing, Furniture and Groceries, rather than being concentrated in only one category.

9.5 Stockout records still contain meaningful sales activity

Revenue was recorded during stockout records across every store and category. This does not mean all of that revenue was lost. It simply shows that the business was still generating sales in periods where the stockout condition was present, making those periods useful for further investigation.

10. Recommendations

Priority	Recommended action	Why it matters
1	Improve stock availability	Review restock frequency and whether stores receive enough stock for expected demand.
2	Prioritize high-opportunity categories	Start with categories showing higher estimated revenue opportunity, especially Toys, Clothing, Furniture and Groceries.
3	Review S004, S005 and S002	Check stock levels, sales patterns and replenishment practices in stores with frequent stockouts.
4	Improve demand planning	Use past sales and demand patterns to improve inventory planning and reduce avoidable demand gaps.
5	Review overstock alongside stockouts	The data shows both high overstock and stockout records, so inventory may not be distributed efficiently across locations/products.

These recommendations are intentionally practical. The analysis points to where FreshMart should investigate first; it does not contain enough information to prescribe exact reorder points, supplier changes or inventory quantities.

11. Limitations and Important Notes

- The dataset source does not specify a currency unit, so monetary values should not be labelled as INR or USD.
- The estimated revenue opportunity is a calculated opportunity, not guaranteed additional revenue.
- The analysis identifies patterns between stockouts, demand gaps and sales, but the dataset alone does not prove causation.
- Demand gap is based on forecast demand minus units sold. It should not automatically be treated as confirmed lost sales.
- Product ID is not unique to one category. There are 20 Product IDs but 100 Product ID + Category combinations, so product-level tables need careful interpretation.
- The dataset source and license have been verified. The source is the Kaggle dataset listed in the Dataset Overview.

12. Conclusion

The analysis suggests that FreshMart's inventory problem is broader than a single product category. Stockouts and demand gaps appear across the business, while a few stores show higher stockout levels than others.

The clearest store-level priorities are S004, S005 and S002 because they have the highest stockout counts. At category level, Toys has the highest demand gap and the highest estimated revenue opportunity in the dashboard.

The main business opportunity is therefore to improve product availability, review stores with repeated stockouts, and use demand patterns more effectively when planning inventory. The project brings these points together through Python analysis, SQL investigation and an interactive Power BI dashboard.

Final project message

The purpose of the project is not simply to report that stockouts exist. It is to show where they are happening, connect them with unmet demand indicators, identify priority areas and turn the findings into actions that can be investigated by the business.

Appendix — Project Workflow

Stage	Output
1. Dataset selection	Retail inventory dataset selected for the business problem
2. Data validation	Quality checks and issue identification
3. Cleaning	Negative demand forecasts handled
4. Feature engineering	Business metrics and flags created
5. Python EDA	Patterns and relationships explored
6. SQL analysis	Business questions answered with SQL
7. Power BI	Three-page interactive dashboard built
8. Findings & recommendations	Business priorities and actions summarized
9. GitHub showcase	Code, dashboard, documentation and presentation organized